

Characterization Log Collection Checklist

To fully characterize the hardware envelope across multiple Cycle Counts (CC), you must perform a "Sweep" for tests 2 through 6. All logs must be captured in **RAW mode**.

WARNING: The order of these tests is strictly hierarchical. Do NOT perform the Saturation or AHRS tests until the others are complete.

1. Calibration Performance (Capture ONLY at 400 CC)

Note: Because calibration matrices are applied to normalized nT values, a single matrix works universally across all Cycle Counts. It must be performed at 400 CC to ensure a high-resolution 50Hz capture of the tumbling sphere.

- ☐ cal_1.csv (Figure-8 calibration sequence)
- ☐ cal_2.csv
- ☐ cal_3.csv
- ☐ cal_4.csv
- ☐ cal_5.csv

2. Noise Floor (Capture at 200, 400, 800, 1600, 3200 CC)

Set the wand on a non-magnetic surface. Do not touch it. Let it log perfectly still for 10 seconds.

- ☐ log_noise_200.csv
- ☐ log_noise_400.csv
- ☐ log_noise_800.csv
- ☐ log_noise_1600.csv
- ☐ log_noise_3200.csv

3. Combined Precision & Isolation (Capture at 200, 400, 3200 CC)

Target: A 10d nail starting 24" away. Lock the wand physically on a table. Protocol: Slide block in 6" increments (1s pause per step). At tip (0"), pause for 3s. Reverse direction in 6" increments (1s pause). Repeat 10x per log.

- ☐ log_repeat_200.csv
- ☐ log_repeat_400.csv
- ☐ log_repeat_3200.csv

4. Saturation Limit (Capture at 200, 400, 3200 CC)

Warning: This test uses a massive magnet and risks permanently magnetizing the wand's soft iron components. Target: Strong magnet staged 10' away. Wand fixed on ground. Protocol: Start at 4' mark. Slide in 6" increments (1s pause per step). At tip (or when UI clips), pause for 3s. Reverse direction in 6" increments (1s pause).

- `[ ] log_saturation_200.csv`
- `[ ] log_saturation_400.csv`
- `[ ] log_saturation_3200.csv`

5. Open Air Rebar Test (Capture at 200, 400, 3200 CC)

Target: 36" rebar starting 8' away. Wand fixed on ground. Protocol: Walk target in 1' increments (1s pause per step). At tip (0'), pause for 3s. Reverse direction in 1' increments. (1 pass per log).

- `[ ] log_rebar_200.csv`
- `[ ] log_rebar_400.csv`
- `[ ] log_rebar_3200.csv`

6. AHRS Stability (Capture at 200, 400 CC)

Warning: Violent swinging can twist the sensor board, invalidating the Kabsch matrix. Point the wand North, aggressively pitch it up and down 45 degrees, and roll it left and right 45 degrees for 10 seconds.

- `[ ] log_ahrs_200.csv`
- `[ ] log_ahrs_400.csv`

Once all logs are collected, process them simultaneously using:

```
python scripts/characterize_system.py \  
--noise "log_noise_200.csv" "log_noise_400.csv" "log_noise_3200.csv" \  
--repeatability "log_repeat_200.csv" "log_repeat_400.csv" "log_repeat_3200.csv" \  
--saturation "log_saturation_200.csv" "log_saturation_400.csv" "log_saturation_3200.csv" \  
--rebar "log_rebar_200.csv" "log_rebar_400.csv" "log_rebar_3200.csv" \  
--ahrs "log_ahrs_200.csv" "log_ahrs_400.csv" \  
--calibration "cal_1.csv" "cal_2.csv" "cal_3.csv" "cal_4.csv" "cal_5.csv"
```